# Algebra Expressions & Equations Cheat Sheet

**Unit Focus:** *simplifying expressions and solving equations*

## 1 KEY VOCABULARY

| | | |
|---|---|---|
| **expression** | A mathematical phrase with numbers, variables, and operations (no equal sign). | $3x - 5$ |
| **term** | A number, variable, or product of a number and variable separated by + or −. | In $4x + 7 - 2y$, the terms are $4x$, $7$, $-2y$. |
| **coefficient** | The numerical factor of a term. | In $5x$, the coefficient is 5. |
| **variable** | A letter that represents a number. | $x$, $y$, $n$, … |
| **constant** | A term with no variable. | 7, −3, 12 |
| **equation** | A statement that two expressions are equal (has an = sign). | $2x + 3 = 11$ |
| **solution** | A value that makes an equation true. | $x = 4$ is a solution of $x + 3 = 7$. |
| **like terms** | Terms with the same variable raised to the same power. | $3x$ and $-5x$ are like terms. |

## 2 ORDER OF OPERATIONS (PEMDAS)

| | | |
|---|---|---|
| **P** | **Parentheses** | Do what's inside ( ) first. |
| **E** | **Exponents** | Evaluate powers. |
| **M** | **Multiply** | Left to right. |
| **D** | **Divide** | Left to right. |
| **A** | **Add** | Left to right. |
| **S** | **Subtract** | Left to right. |

When operations are on the same level, work from left to right!

**Example:** Evaluate $3 + 4(2^2 - 5)$.

**Step 1: Parentheses:** $2^2 - 5 = 4 - 5 = -1$ $\quad 3 + 4(-1)$

**Step 2: Multiply:** $3 + 4(-1) = 3 + (-4)$

**Step 3: Add:** $3 + (-4) = -1$

**Answer: −1**

## 3 EVALUATING EXPRESSIONS

To evaluate an expression, substitute the given value(s) for the variable(s) and simplify.

**Example:** Evaluate $2x + 5$ when $x = 3$.

**Step 1: Substitute:** $2(3) + 5$

**Step 2: Multiply:** $6 + 5$

**Step 3: Simplify:** $11$

**Answer: 11**

## 4 COMBINING LIKE TERMS

Add or subtract the coefficients of like terms (terms with the same variable and exponent).

**Example:** Simplify $4x - 7 + 2x + 5 - 3x$.

**Step 1: Group like terms:** $(4x + 2x - 3x) + (-7 + 5)$

**Step 2: Combine coefficients:** $(4 + 2 - 3)x + (-2)$

**Step 3: Simplify:** $3x - 2$

**Answer: 3x − 2**

## 5 DISTRIBUTIVE PROPERTY

The distributive property lets you multiply a number (or variable) by each term inside parentheses.

$$a(b + c) = ab + ac$$

**Example:** Simplify $3(x - 4)$.

**Step 1:** Distribute 3 to each term:

$$3(x - 4) = 3(x) + 3(-4)$$

**Step 2: Simplify:** $3x - 12$

**Answer: 3x − 12**

## 6 SOLVING EQUATIONS

Use inverse operations to isolate the variable.

### One-Step Equations

**Example:** Solve $x + 6 = 14$.

**Step 1:** Subtract 6 from both sides.

$x + 6 - 6 = 14 - 6$

$x = 8$

**Answer: x = 8**

### Two-Step Equations

**Example:** Solve $3x - 5 = 16$.

**Step 1:** Add 5 to both sides.

$3x - 5 + 5 = 16 + 5$

$3x = 21$

**Step 2:** Divide by 3.

$\frac{3x}{3} = \frac{21}{3}$

$x = 7$

**Answer: x = 7**

## 7 QUICK REMINDERS & COMMON MISTAKES

### DO REMEMBER:

- Follow the order of operations (PEMDAS).
- Combine only like terms.
- Distribute to every term inside parentheses.
- Use inverse operations to solve equations.
- Check your solution by substituting it back into the equation.

### COMMON MISTAKES:

- Forgetting to follow PEMDAS. *(Do parentheses and exponents first!)*
- Combining unlike terms. $3x + 5 \neq 8x$
- Not distributing to all terms. $3(x + 4) \neq 3x + 4$ *(it is $3x + 12$)*
- Moving a term to the other side without changing the sign. $x - 5 = 9 \rightarrow x = 9 + 5$ *(not $9 - 5$)*
- Forgetting to check your solution.

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# Algebra Expressions & Equations Cheat Sheet

**More Practice:** *solving equations, proportions, and literal equations*

## 1 WORDS TO ALGEBRA

Common words and phrases used to write expressions and equations.

| Words/Phrases | Algebra | Words/Phrases | Algebra |
|---|---|---|---|
| sum, total, plus | $a + b$ | increased by | $a + b$ |
| difference, minus | $a - b$ | decreased by | $a - b$ |
| product, times | $ab$ | twice a number | $2x$ |
| quotient, divided by | $\frac{a}{b}$ | three times a number | $3x$ |
| a number | $x$ | half of a number | $\frac{x}{2}$ |
| a variable | $x, y, n$ | a number squared | $x^2$ |
| is, equals | $=$ | a number cubed | $x^3$ |
| is not equal to | $\neq$ | the absolute value of a number | $\lvert x \rvert$ |

## 2 WRITE EXPRESSIONS & EQUATIONS

Translate each verbal phrase into an algebraic expression or equation.

**Examples:**

a) Seven more than a number. $x + 7$

b) The product of 3 and a number. $3x$

c) A number decreased by 5. $x - 5$

d) Twice a number is 12. $2x = 12$

e) The sum of a number and 4 is 20. $x + 4 = 20$

f) Three times a number minus 6. $3x - 6$

g) Half of a number is equal to 8. $\frac{x}{2} = 8$

Look for key words like sum, difference, product, quotient, is, and more!

## 3 SOLVING MULTI-STEP EQUATIONS

Use inverse operations and follow the order of operations. Simplify both sides step by step.

**Example:** Solve $3(x + 2) - 5 = 16$.

**Step 1:** Distribute: $3x + 6 - 5 = 16$

**Step 2:** Combine like terms: $3x + 1 = 16$

**Step 3:** Subtract 1: $3x = 15$

**Step 4:** Divide by 3: $x = 5$

**Answer:** $x = 5$

## 4 VARIABLES ON BOTH SIDES

Collect like terms on the same side.

**Example:** Solve $4x - 3 = 2x + 9$.

**Step 1:** Subtract $2x$ from both sides: $4x - 3 - 2x = 2x + 9 - 2x$

$2x - 3 = 9$

**Step 2:** Add 3 to both sides: $2x = 12$

**Step 3:** Divide by 2: $x = 6$

**Answer:** $x = 6$

## 5 CHECKING A SOLUTION

Substitute the value back into the original equation.

**Example:** Check if $x = 4$ is a solution of $2x - 5 = 3$.

**Step 1:** Substitute $x = 4$.

$2(4) - 5 = 3$

$8 - 5 = 3$

$3 = 3$ ✓

**Step 2:** Compare both sides.

Since both sides are equal, $x = 4$ is a solution.

## 6 SOLVING PROPORTIONS

Use cross products to solve for the unknown.

**Example:** Solve $\frac{3}{4} = \frac{x}{12}$.

**Step 1:** Cross multiply:

$3(12) = 4(x)$

$36 = 4x$

**Step 2:** Divide by 4:

$x = 9$

**Answer:** $x = 9$

For a proportion $\frac{a}{b} = \frac{c}{d}$, use cross products: $ad = bc$.

## 7 LITERAL EQUATIONS (FORMULAS)

Solve a formula for a different variable.

**Example:** Solve $A = \frac{1}{2}bh$ for $h$.

**Step 1:** Multiply both sides by 2:

$2A = bh$

**Step 2:** Divide by $b$:

$h = \frac{2A}{b}$

**Answer:** $h = \frac{2A}{b}$

Treat other variables as constants. Use inverse operations to isolate the desired variable.

## 8 SPECIAL CASES & QUICK REMINDERS

**No Solution (Contradiction)**

**Example:** $3(x + 2) = 3x + 9$

$3x + 6 = 3x + 9$

Subtract $3x$: $6 = 9$

No solution (since $6 \neq 9$).

**Infinitely Many Solutions (Identity)**

**Example:** $4(x - 3) = 4x - 12$

$4x - 12 = 4x - 12$

Infinitely many solutions (same expression on both sides).

**Quick Reminders:**

- Keep the equation balanced (do the same thing to both sides).
- Combine like terms after distributing.
- Watch for special cases: no solution or infinitely many solutions.
- Check your solution by substitution.

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